

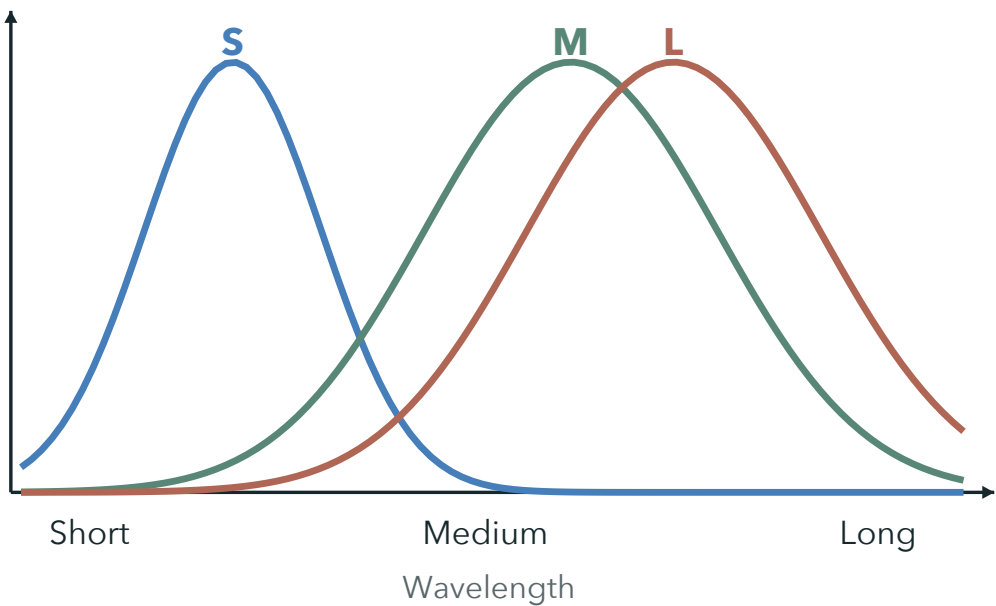
Colour channels

Three overlapping cone classes feed neural comparisons.

Pigment classes are not colour labels

A | Overlapping S, M and L sensitivities

Relative response



Curves show relative overlap only. Their heights, widths and wavelength positions are schematic.

Class versus functional loss

Class	Loss of an independent class	Shifted class sensitivity
L: longer wavelengths	Protanopia	Protanomaly
M: medium wavelengths	Deutanopia	Deuteranomaly
S: shorter wavelengths	Tritanopia	Tritanomaly

Fast check

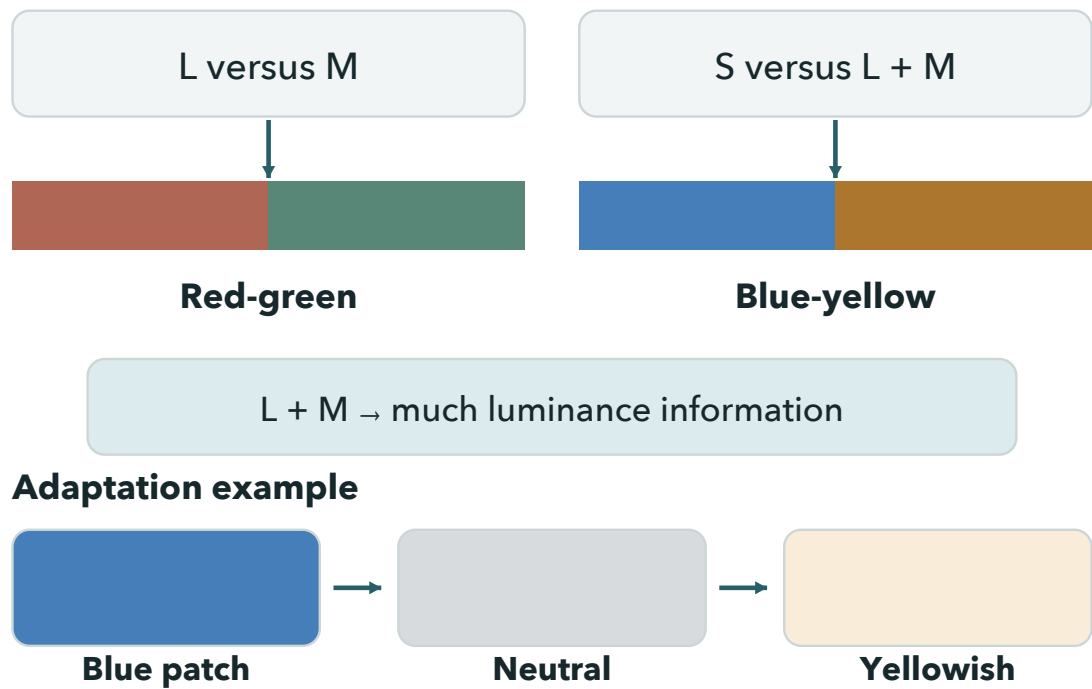
A cone’s response alone cannot distinguish wavelength from intensity. Colour requires comparison across classes; L and M sensitivities overlap substantially.

Figure reference

A: Webvision colour vision and photoreceptors. Original schematic.

Opponent comparisons and afterimages

B | From cone input to opponent signals



These are simplified functional comparisons, not one universal equation for every retinal neuron.

Useful channels

Comparison	Information emphasised
L versus M	Red-green chromatic differences
S versus combined L + M	Blue-yellow chromatic differences
Combined L + M	Much of photopic luminance information

Afterimage check

After blue adaptation, neutral input can look yellowish as opponent balance shifts. Adaptation can involve several processing levels; it does not require a fourth “yellow cone”.

Figure reference

B: Webvision colour vision. Original comparison and afterimage schematic.